SURI Machine Learning Data Requirements Matrix V2

To build a robust machine learning model for SURI, the Department of Agriculture Region 7 soils laboratory needs a clear specification of both the input variables (the soil test results) and the output targets (the fertilizer recommendations). The data must be provided as paired records so that the model can successfully learn how specific soil chemical properties and target crops dictate the required nutrient applications.
This updated matrix includes the comprehensive set of standard and specialized parameters available from the laboratory, incorporating qualitative remarks, geospatial data, and temporal features.

1. Core Routine Set (Standard DA RSL Dataset)

Data Category	Parameter / Field	Unit	Parameter Explanation	ML Training Use Case
Metadata	Laboratory Code	Alphanumeric	The unique identifier assigned by the DA Region 7 lab.	**Primary Key:** Securely executes SQL joins pairing inputs with targets.
Metadata	Sampling Site / Location	Text	Geographical origin of the sample (e.g., Jampang, Argao).	**Geospatial Feature:** Links soil profiles to regional polygon structures in

Data Category	Parameter / Field	Unit	Parameter Explanation	ML Training Use Case
				QGIS.
Metadata	Date of Sampling	Date	The day the sample was extracted.	**Temporal Feature:** Accounts for seasonal nutrient shifts and wash-outs.
Metadata	Crop to be Planted	Text	Targeted plant species (e.g., Sorghum).	**Categorical Feature:** Dictates baseline nutrient requirements.
Metadata	Farm Area	Sq. M. / ha	Total land area evaluated.	**Scaling Feature:** Normalizes target fertilizer volumes.
Input Feature	pH (H_2O, 1:1)	Unitless	Soil acidity or alkalinity balance.	**Numerical Feature:** Determines nutrient

Data Category	Parameter / Field	Unit	Parameter Explanation	ML Training Use Case
				availability thresholds.
Input Feature	Organic Matter (OM)	%	Percentage of decomposed material.	**Numerical Feature:** Gauges natural soil fertility.
Input Feature	Available Phosphorus (P)	ppm	Amount of immediately accessible P.	**Numerical Feature:** Dictates P output target.
Input Feature	Exchangeable Potassium (K)	ppm	Amount of readily available K.	**Numerical Feature:** Dictates K output target.
Input Feature	Total Nitrogen (N)	%	Total nitrogen reserve in soil profile.	**Numerical Feature:** Dictates N output target.

Data Category	Parameter / Field	Unit	Parameter Explanation	ML Training Use Case
Qualitative Input	Lab Remarks (pH, OM, P, K)	Text	Expert classifications (e.g., "Moderately Alkaline", "Sufficient+").	**Categorical Feature:** Provides model with pre-classified agronomic thresholds.
Target (Label)	Nutrient Req (N, P_2O_5, K_2O)	kg/ha	Elemental weight of N, P, K required.	**Regression Output:** Primary numerical prediction targets.
Target (Label)	Application 1 & 2 Formulations	kg/ha	NPK bag blend and timing (e.g., 14-14-14 at planting).	**Multi-class Target:** Recommends standard commercial blends and timing.

2. Extended Capabilities (Specialized & Physical Parameters)

These optional parameters from the BSWM Soil Analysis Request Form can be added as supplementary features if specialized datasets are available.

Parameter Group	Included Features	ML Training Use Case
Physical Parameters	Texture, Sand Fraction, Water Holding Capacity (WHC), Field Capacity (FC), Permanent Wilting Point (PWP), Moisture Content, Bulk Density	Improves predictions related to water retention and physical soil structure impact on roots.
Salinity & Alkalinity	Electrical Conductivity (EC), Cations (Ca, Mg, Na, K), Anions (CO_3, HCO_3, Cl, SO_4), Sodium Adsorption Ratio (SAR)	Critical for tuning recommendations in saline or coastal agricultural areas.
Specialized & Micronutrients	pH ($CaCl_2$), Exchangeable Acidity, Cation Exchange Capacity (CEC), Micronutrients (Cu, Zn, Mn, Fe), Microbial Count	Allows for advanced agronomic modeling and highly specialized micronutrient prescriptions.

3. Important Notes for the Laboratory Request

- **Data Structure:** The "Soil Chemical Test Report" and the "Fertilizer Recommendation Report" must be strictly paired using the **Laboratory Code** (e.g., SCC 417). An orphaned soil test without its corresponding recommendation cannot be used to train the model.
- **Format:** Request the historical data export in a flat, tabular format (like CSV or Excel) rather than raw PDFs. This will drastically speed up the data ingestion pipeline into the PostgreSQL database.
- **Test Methods:** Ensure the lab maintains consistent testing methods across all provided samples (such as Potentiometric for pH and Olsen/Bray 1 for Available Phosphorus) to prevent data drift or scaling issues during model training.